```
#include<stdio.h>
#include<stdlib.h>
struct node {
	int data;
	struct node *next;
};
struct node *top=NULL,*temp,*newNode;

void push(int data)
{
	newNode = (struct node*)malloc(sizeof(struct node));
	newNode->data = data;
	newNode->next = NULL;

	if(top == NULL)
	{
		top = newNode;
	}else{
		newNode->next = top;
		top = newNode;
	}
	printf("element pushed:%d \n",data);
}

int pop()
{
	if (top == NULL) {
		printf("\nEMPTY STACK");
	} else {
		temp = top;
		int temp_data = top->data; //to store data of top node
		top = top->next;
		free(temp); //deleting the node
		return temp_data;
	}
}

void display() {
	// Display the elements of the stack
	if (top == NULL) {
		printf("\nStack Underflow\n");
	} else {
		printf("\n The stack is: \n");
		temp = top;
		while (temp != NULL) {
			printf("%d--->", temp->data);
			temp = temp->next;
		}
		printf("NULL\n\n");
	}
}

int main()
{
	push(10);
	push(20);
	push(30);
	display();
	printf("\nelement poped %d",pop());
	printf("\nelement poped %d",pop());
```

```
    display();
    return 0;
}
```